

MECH 3620 Checklist for Final Project Submission

1 Project's summary (10 Points)

1. Executive summary (1 page) (**3 points**)
2. Dimensioned 3-view drawings and perspective for your final design. They should be on the same scale, align with each other, and be on the same page. Show the following information on the drawing (**5 points**):
 - ☐ CG location and excursion diagram
 - ☐ Neutral point location
 - ☐ Control surfaces and high-lift devices
 - ☐ Tip-over and take-off rotation lines
 - ☐ Landing gears
3. Summary table (1 page) (**2 points**):
 - ☐ OEW, Payload, MTOW
 - ☐ Cruise, landing, and take-off C_L
 - ☐ Cruise L/D
 - ☐ T/W
 - ☐ W/S
 - ☐ Engine type, model, and maximum sea-level-static thrust
 - ☐ Wing geometry: b , S , AR, avg t/c
 - ☐ Cruise Mach number
 - ☐ Static margin
 - ☐ Maximum range (nmi)
 - ☐ Number of passengers
 - ☐ Maximum take-off and landing distance
 - ☐ Maximum R/C and climb angle

2 Main Content (90 points)

This checklist serves as a minimum requirement. Don't forget to discuss the design rationales!

1. Introduction and sizing (**8 points**)
 - ☐ Aircraft requirements and mission profile
 - ☐ Matching chart and design point
 - ☐ Weight estimation and sizing results
 - ☐ New technology: what makes your aircraft unique in the market (including measures of merit)
2. Interior layout (**7 points**)
 - ☐ Three-view drawings with dimensions showing cockpit, cabin, etc
3. Aerodynamics (**15 points**)
 - ☐ Wing design
 - ☐ Airfoil selection
 - ☐ Stall speed

4. Weights and balance (**8 points**)
 - ☐ Component breakdown & location
 - ☐ Weight estimation
 - ☐ CG range estimation
 5. Stability and control (**15 points**)
 - ☐ Static stability calculation
 - ☐ Empennage design & sizing
 6. Structures (**10 points**)
 - ☐ Structural design concept
 - ☐ V-n diagram
 - ☐ Distribution of air loads on the wings
 7. Landing gear design (**5 points**)
 - ☐ Landing gear concept
 - ☐ Disposition & sizing
 8. Propulsion system (**5 points**)
 - ☐ Engine selection
 - ☐ Engine integration
 9. Cost analysis [Optional] (**4 bonus points**)
 - ☐ Cost estimation
 - ☐ Direct operating cost (DOC)
 - ☐ Break even point (BEP), detailed calculation and assumptions involved
 10. Discussions, comments, opinions on the refinements and results (**12 points**)
 - ☐ Did you do any further attempts to refine your design (e.g., sizing matrix, sensitivity studies, optimization, or other analyses? Please elaborate.
 - ☐ After refinement, what are the changes? Is the aircraft performance better and compliant to all requirements?
 - ☐ Anything you can think to improve the design/configuration if you have extra time and resources?
 11. References
 12. Appendices
 - ☐ Supplementary engineering drawings (if any)
 - ☐ Details of analysis (if any)
- ☐ Pay special attention to describing methods that you used but were not covered in class, and reference all sources
- ☐ Ensure document format, numbering, captioning, referencing, etc. are neat and consistent! (**5 points**)

3 Individual Part

For this individual part, there will be a separate submission portal on Canvas (i.e., NOT together with the final project report and each student will submit it individually), and it is due on the same day as the final report (15 May 2026). The individual part should include the following (please refer to the [Project Description](#) for more information):

1. Statement of contribution (**10 points**)
 - ☐ Analyses and calculations that you did
 - ☐ Programming contribution
 - ☐ Report writing contribution
 - ☐ Additional contributions/achievements (optional)
2. Tradeoff analysis (**40 points**)
 - ☐ Details and background of the trade-off study
 - ☐ Calculation, observation and analysis
 - ☐ Importance of trade-off analysis
3. Optimization problem formulation (**30 points**)
 - ☐ Problem definition
 - ☐ Objective function, constraints, and decision variables
 - ☐ Proposed methodology
4. Self-reflection on learning experience (**20 points**)
 - ☐ Your course experience
 - ☐ Your current perspective toward “aircraft design”
 - ☐ The most interesting/inspiring/insightful thing(s) you learn from this course

The peer assessment will be submitted separately and confidentially.

Please adhere to project guidelines and fulfill report requirements. Make sure you tick all the boxes in the checklist – if you can’t complete some items, explain why. Failing to do so might make you lose some marks.

Project grading schemes (please refer to the [Project Description](#) for more information):

- The final project score will be individually different than your group score.
- The peer assessment score corresponds to 30% of the Individual’s project score.

$$IS = GS(0.7 + 0.3 * PA) \quad (1)$$

Where:

IS: Individual final project score

GS: Group score

PA: Peer assessment score